

Problem Set 2 – Set identities

Session 2 · Sets & set operations

Part A · Notation and computation

1 NOTATION

Let $U = \{1, 2, \dots, 12\}$, $A = \{x \in U : x \text{ is even}\}$ and $B = \{x \in U : x \text{ is a multiple of } 3\}$.

- Write A and B in roster notation.
- Compute $A \cap B$, $A \cup B$, $A \setminus B$, $A \triangle B$ and B^c .

2 $\in$ VS $\subseteq$

Let $C = \{1, \{1\}, \{1, 2\}, \emptyset\}$. Decide true or false, and in each case quote the definition that settles it.

- $1 \in C$
- $\{1\} \in C$
- $\{1\} \subseteq C$
- $\{1, 2\} \subseteq C$
- $\emptyset \subseteq C$
- $|C| = 4$

3 POWER SET

Let $A = \{a, b, c\}$.

- List $\mathcal{P}(A)$ in full, organised by the size of the subset.
- State $|\mathcal{P}(A)|$ and explain in one sentence why $|\mathcal{P}(A)| = 2^{|A|}$ for *any* finite A .

4 SET-BUILDER

Let $S = \{x \in \mathbb{Z} : x^2 < 10\}$.

- Write S in roster notation, and say in one sentence how you know the list stops where it does.
- Write $\{2, 4, 6, 8\}$ in set-builder notation.
- Is $-3 \in S$? Is $\{0, 1\} \subseteq S$? Justify each from the definition.

5 THE EMPTY SET

Decide true or false, and in each case name the definition that settles it.

- a. $\emptyset \in \{\emptyset\}$
- b. $\emptyset \subseteq \{\emptyset\}$
- c. $|\{\emptyset\}| = 1$
- d. $\emptyset = \{\emptyset\}$
- e. $\mathcal{P}(\emptyset) = \{\emptyset\}$

Part B · Six identities to prove

6 DISTRIBUTIVITY

Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for all sets A, B, C .

7 DE MORGAN

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$.

8 DIFFERENCE

Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ for all sets A, B, C .

9 SYMMETRIC DIFFERENCE

Prove that $A \triangle B = (A \cup B) \setminus (A \cap B)$ for all sets A, B .

10 SUBSETS

Prove that for all sets A, B : $A \subseteq B \iff A \cup B = B$.

11 CARTESIAN PRODUCT

Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$ for all sets A, B, C .

Part C · Counterexamples and writing

12 DISPROVE

Each claim below is false in general. For each one: name explicit finite sets, compute **both** sides, and point at an element that is in one side and not the other.

- a. $A \setminus (B \setminus C) = (A \setminus B) \setminus C$
- b. $A \times B = B \times A$
- c. $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$

A student hands in the following.

Claim. $A \cap (B \cup C) = (A \cap B) \cup C$ for all sets A, B, C .

Proof. Draw three overlapping circles. Shade $A \cap (B \cup C)$, then shade $(A \cap B) \cup C$. Both shadings cover the region in the middle where all three circles meet, so the two sets are equal. ■

- Which step of this argument fails, and why? Answer in one or two sentences.
- Give a counterexample to the claim.